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Location: 20143 Milan, Italy



Alessandro Pedone

Curriculum Vitae

► PERSONAL DATA

Date of Birth	01/10/2002
Nationality	Italian
Place of Birth	Milan, Italy

► CURRENT STATUS

Politecnico di Milano	Pursuing Master's Degree in Mathematical Engineering, track in Computational Science and Computational Learning
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► EDUCATION

2024 - Current	Master's Degree in Mathematical Engineering, Politecnico di Milano
2021 - 2024	Bachelor's Degree in Mathematical Engineering, Politecnico di Milano Final grade: 110/110 cum laude Thesis: The Cauchy-Kowalevski theorem and some of its consequences Keywords: PDEs, characteristics method, analyticity/holomorphy, power series, method of majorants, Cauchy-Kowalevski, Holmgren and Cartan-Kähler theorems Supervisor: Prof. Maurizio Grasselli

► EXPERIENCE

14/09/2026 - 20/11/2026	Master's Thesis, SISSA
29/06/2026 - 03/07/2026	GMT: unraveling the structure of singularities, UniTrento
01/12/2025 - 03/12/2025	Junior Math Days 2025, SISSA
22/09/2025 - 26/09/2025	Workshop and Summer School on Applied Analysis 2025, TU Chemnitz
23/06/2025 - 27/06/2025	Theoretical Foundations of Machine Learning 2025, UniGe

► PROJECTS

04/2025 - 09/2026	deeponet-for-mems Description: Coupling a DL-ROM with a traditional ROM to solve multi-physics problem (MEMS simple model).
03/2026	Python Workflow Guide  Description: comprehensive guide for building a Python workflow for (scientific computing) projects, mainly aimed at Windows and Linux users with limited terminal experience.

03/2025 - 05/2025

C++ Scientific Computing Projects

1. Some optimization methods for real-valued functions (Gradient Descent, Heavy Ball, Nesterov, ADAM...)
2. Parallel implementations of two kind of solvers for the Poisson equation in the unit square (Schwarz method)
3. Implementation of a matrix class, particularly suited for sparse matrices (CSR/CSC formats and modified formats)

11/2024 - 12/2024

Deep Learning Projects

1. Mars Terrain segmentation of grayscale images (UNet architecture).
2. Blood Cell Classification of RGB images (CNNs and Transfer Learning).

04/2024 - 06/2024

Statistical Inference for Math Performance

Description: a linear regression model for math performance of italian high school students trained on the OECD PISA 2022 survey.

► **SKILLS**

Programming Languages

Python, C/C++, MATLAB, R, LaTeX

Libraries

NumPy, SciPy, scikit-learn, TensorFlow, Keras, Pandas, Matplotlib, Seaborn, MIP, FEniCS, FEniCSx

Operating Systems

Linux (Ubuntu), Windows

Additional Software

Notion

Memory techniques

Method of loci, Mnemonic major system, Mnemonic link system

► **LANGUAGES**

Italian

Native speaker

English

C1 (IELTS 7.5)

► HONOURS AND AWARDS

25/01/2023

Premio Migliori Matricole aa 21-22 Classe L 8 Fondo Giovani

Best Freshmen Award, academic year 2021-2022, Class L-8, Fondo Giovani — granted upon completion of the first year of enrollment, based on academic performance, including GPA and the number of ECTS credits earned.

► MEMBERSHIPS

2025 - Current

SIAM (Society for Industrial and Applied Mathematics)

2021 - Current

AIM (Associazione Ingegneri Matematici)

► VOLUNTEERING

2020 - Current

Blood donor

2018

ABCDigital: Liceo Scientifico Elio Vittorini promoted a project to organize a free course for senior citizens, consisting of 10 weekly lectures (2 hours each), aimed at improving their digital skills, device usage, and online awareness.